

Question

A is a monoprotic weak acid that is liquid at room temperature, which reacts with sodium hydroxide to obtain salt **B**. **A** reacts with sodium borohydride to obtain salt **D**, while **B** reacts with a chloride **C** of a short-period main group element **M** to obtain salt **E**. It is known that the mass fractions of nitrogen in **D** and **E** are both greater than 80%.

A fluoride **F** of another short-period main group element reacts with antimony pentafluoride in liquid hydrogen fluoride to obtain **G**. The mass fraction of Sb in **G** is 43.06%. **G** reacts with **A** in liquid hydrogen fluoride to obtain an unstable high-energy material **H**. **H** reacts with **D** and **E** in liquid sulfur dioxide to obtain **I** and **J** respectively. The cations of these two substances are the same, and the mass fractions of nitrogen in the anions are 95.66% and 91.23% respectively, and the stability of **I** and **J** is lower than that of **H**.

From the above, the following options are correct:

- A.** The chemical formula of **D** contains only two elements.
- B.** The structure of **C** is a trigonal pyramidal structure.
- C.** The mass fraction of nitrogen in **E** is approximately 85%
- D.** **F** has four conformational isomers.
- E.** There are no π bonds in the cation of **G**.
- F.** The mass fraction of nitrogen in **H** is approximately 27%.
- G.** The cations of **I,J** have a linear structure.
- H.** **J** and **E** have different anions.

- I. All of the above options are incorrect.

Answer

Correct Answer: **I**

Detailed Explanation

First, the question states that nitrogen element exists in **D**, which can only come from **A**; the only monobasic liquid weak acids containing nitrogen are HCN , HN_3 , HNO_2 . Since the mass fraction of nitrogen in **D** is greater than 80%, **A** can only be HN_3 to satisfy this mass fraction.

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1 PTS

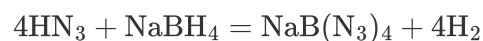
The only monobasic liquid weak acids containing nitrogen are HCN , HN_3 , HNO_2

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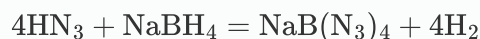
A can only be HN_3 to satisfy the mass fraction.

HN_3 reacts with sodium hydroxide to produce **B**, sodium azide, and reacts with sodium borohydride to obtain **D**. Since the nitrogen content of **D** is extremely high, the azide radical will certainly not be reduced. Thus, hydrogen ions are reduced by negative hydrogen, and the azide radical coordinates with the boron atom. The apparent result of the reaction is that negative hydrogen is replaced by the azide radical:



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D is $\text{NaB}(\text{N}_3)_4$, containing three elements, so option A is incorrect.

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D is $\text{NaB}(\text{N}_3)_4$, containing three elements, so option A is incorrect.

Currently, there is too little information about **E** to deduce, so let's continue to deduce further:

The fluoride **F** of a short-period main group element reacts with antimony pentafluoride in liquid hydrogen fluoride to obtain **G**. The mass fraction of Sb in **G** is 43.06%. Antimony pentafluoride, as a strong Lewis acid, readily forms SbF_6^- anions in liquid hydrogen fluoride. Considering that **G** contains one SbF_6^- , the relative molecular mass of the remaining cation is

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Antimony pentafluoride readily forms SbF_6^- in liquid hydrogen fluoride

$$121.8/0.4306 - 121.8 - 19 \times 6 = 47$$

Since **F** is a fluoride, consider that the cation contains fluorine atoms:

If it contains one fluorine atom, the remaining mass is 28, which is exactly two nitrogen elements. If it contains two fluorine atoms, the remaining mass is 9, which corresponds to no element. Therefore, the chemical formula of the cation of **G** is N_2F^+ , and the chemical formula of **G** is $[\text{N}_2\text{F}^+][\text{SbF}_6^-]$.

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The chemical formula of the cation of **G** is N_2F^+ , and the chemical formula of **G** is $[\text{N}_2\text{F}^+][\text{SbF}_6^-]$.

In N_2F^+ , in order to satisfy the charge, a double bond needs to be formed between the nitrogen elements, thus there is a π bond, and option E is incorrect.

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1 PTS

N_2F^+ has a π bond, and option E is incorrect

F is a fluoride and N_2F^+ is generated, so its structure is obviously N_2F_2 . To satisfy the Lewis structure, its bonding mode can be expressed as $\text{F} - \text{N} = \text{N} - \text{F}$, which has cis and trans isomers, so option D is incorrect.

CHECKPOINT

1 PTS

F is N_2F_2 , which has cis and trans isomers, so option D is incorrect

$[\text{N}_2\text{F}^+][\text{SbF}_6^-]$ reacts with hydrazoic acid in liquid hydrogen fluoride to generate **H**. Since the anion is relatively stable, it should not change; therefore, consider the reaction of the cation. High-energy materials easily lead to the idea that the cation contains an azide radical structure, and then infer that the cation is N_3^- replacing the fluorine of N_2F^+ , generating N_5^+ ions, so the preliminary inference is that **H** is $[\text{N}_5^+][\text{SbF}_6^-]$, and its correctness remains to be verified below.

CHECKPOINT

1 PTS

The anion is relatively stable, so it should not change

CHECKPOINT

1 PTS

The cation may be N_5^+ ions

Further infer that **H** reacts with **D**, i.e., $\text{NaB}(\text{N}_3)_4$, to produce **I** with a nitrogen content of 95.66% and lower stability. Thus, **I** has a large number of nitrogen atoms, and the structure of $\text{B}(\text{N}_3)_4^-$ should be retained;

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The structure of $\text{B}(\text{N}_3)_4^-$ should be retained

If the N_5^+ ion inferred above replaces the sodium ion, the nitrogen content of the generated $[\text{N}_5^+][\text{B}(\text{N}_3)_4^-]$ is exactly 95.66%, and the substance is more unstable than **H** due to its excessive nitrogen content, which is in line with the meaning of the question. Thus, substance **I** is $[\text{N}_5^+][\text{B}(\text{N}_3)_4^-]$. Therefore, the chemical formula of **H** is also verified to be correct.

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2 PTS

Substance **I** is $[\text{N}_5^+][\text{B}(\text{N}_3)_4^-]$

CHECKPOINT

1 PTS

The chemical formula of **H** is $[\text{N}_5^+][\text{SbF}_6^-]$, and is verified to be correct

The mass fraction of nitrogen in $[\text{N}_5^+][\text{SbF}_6^-]$ is 22.9%, so option F is incorrect.

CHECKPOINT

1 PTS

The mass fraction of nitrogen in $[\text{N}_5^+][\text{SbF}_6^-]$ is 22.9%, so option F is incorrect.

At this point, let's go back and infer the chemical formulas of **E** and **J**. First of all, the cation of **J** is also $[\text{N}_5^+]$. At the same time, to satisfy the nitrogen content of 91.23%, the anion must contain nitrogen.

It is known that **E** is generated by the reaction of sodium azide with a chloride of a short-period main group element **M**, and the nitrogen content in **E** is also greater than 80%. Therefore, there should also be multiple azide radicals similar to **D** in **E**, and the cation is a sodium ion. The reaction of **D** and **H** should be similar to that of **E**, which is a double decomposition reaction, that is, the anion of **J** is the same as **E**, so option H is incorrect.

CHECKPOINT

1 PTS

There should also be multiple azide radicals similar to **D** in **E**, and the cation is a sodium ion.

CHECKPOINT

1 PTS

The anion of **J** is the same as **E**, so option H is incorrect

Therefore, the chemical formula of **I** can be set as $[\text{N}_5^+][\text{M}(\text{N}_3)_m^-]$, where according to the question, **M** is a short-period main group element. Iterate through the values of m :

When m is 1, the remaining relative molecular mass is $(5 + 3 \times 1) \times 14 / 0.9123 - (5 + 3 \times 1) \times 14 = 10.8$, which corresponds to no element.

When m is 2, with the same method, the remaining relative molecular mass is 14.8; when m is 3, the remaining is 18.8; when m is 4, the remaining is 22.9; when m is 5, the remaining is 26.9, all of which correspond to no element.

When m is 6, the remaining is 31, which is exactly the phosphorus element, and according to the valence, the phosphorus element is +5, which also conforms to the element properties. Therefore, **M** is the phosphorus element.

CHECKPOINT

2 PTS

M is the phosphorus element, $m=6$

Therefore, the common anion of **J** and **E** is $[\text{P}(\text{N}_3)_6^-]$, the chemical formula of **E** is $[\text{Na}^+][\text{P}(\text{N}_3)_6^-]$, and the chemical formula of **J** is $[\text{N}_5^+][\text{P}(\text{N}_3)_6^-]$.

CHECKPOINT

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The chemical formula of **E** is $[\text{Na}^+][\text{P}(\text{N}_3)_6^-]$, and the chemical formula of **J** is $[\text{N}_5^+][\text{P}(\text{N}_3)_6^-]$

The mass fraction of nitrogen in $[\text{Na}^+][\text{P}(\text{N}_3)_6^-]$ is 82.4%, so option C is incorrect.

CHECKPOINT

1 PTS

The mass fraction of nitrogen in $[\text{Na}^+][\text{P}(\text{N}_3)_6]^-$ is 82.4%, so option C is incorrect.

Since M is the phosphorus element and its valence is +5, its chloride C is obviously phosphorus pentachloride, which has a trigonal bipyramidal structure, so option B is incorrect.

CHECKPOINT

1 PTS

C is phosphorus pentachloride, which has a trigonal bipyramidal structure, so option B is incorrect.

The cations of I and J are both $[\text{N}_5^+]$, and its structure is, according to the above inference, the azide radical replacing $[\text{N}_2\text{F}^+]$, so it can be understood as two nitrogen-nitrogen triple bonds connected by a nitrogen atom as a bridging group. The bridging nitrogen atom is sp^3 hybridized, so the entire ion structure is bent, and option G is incorrect.

CHECKPOINT

1 PTS

I and J cations are both $[\text{N}_5^+]$, which is bent, and option G is incorrect

In summary, all options A-H are incorrect, so option I of this question is correct.

CHECKPOINT

1 PTS

All options A-H are incorrect, so option I of this question is correct

